Brownian sticky clocks: construction, holding time and causal recovery in the observed filtration

Project Stardust
A companion to a frozen verified library

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Abstract

We reconstruct the complete construction of a Brownian sticky time change from smooth convex approximations to absolute value. A compact probability estimate and a summable subsequence give a measurable continuous increasing mass, whose Stieltjes measure is supported on Brownian zeros. Its inverse clock has an exact physical holding-time identity. The clock is recoverable from the observed path up to the current time, so both the process and its compensated square are martingales in explicitly specified observed filtrations. At every positive parameter and deterministic horizon, expected holding is positive and the second moment is strictly smaller than elapsed time. An independent Brownian source on the complementary clock yields two genuine Brownian coordinate processes whose covariance equals expected holding. Their finite-increment laws are proved by stopped Fourier products, rather than inferred from Gaussian one-time marginals. These classical mechanisms are presented with full proofs and exact formal-source correspondences. The construction does not use the separately unfinished closed-generator or stochastic-flow identification. Operational-time Hermite and explicit sticky boundary martingales are proved in the observed filtration, with the derivative-jump factor fixed. An explicit vanishing boundary pair has generator value $1/4$ at zero, showing that uniformly dense flat smooth tests are not a graph core. Borel maps recover the complete Wiener and sticky paths from one another almost surely, yielding an onto whole-path L^2 isometry and exact transport of supplied closed operator and adjoint domains. The complete spatial occupation measure retains its holding atom; current-state conditional operators have an exact composition residual.

1 The construction and its precise conclusions

Let $B = (B_s)_{s\geq 0}$ be standard real Brownian motion, $B_0 = 0$, with a measurable realization in $C([0,\infty),\mathbb{R})$. We use its raw natural filtration, augmented by ambient measurable null events:

$$\mathcal{F}_s = \sigma(B_u : 0 \leq u \leq s) \vee \mathcal{N}.$$

Here $\mathcal{N}$ consists of the null events in the ambient probability space; this convention suffices for all modifications below. No right-continuous enlargement of the filtration is implicit. The Brownian stochastic integral is its ordinary predictable $L^2(ds\,d\mathbb{P})$ isometry, with conditional expectations at deterministic horizons. These standard Brownian foundations are also constructed in the frozen source library.

Theorem 1.1. *There is a measurable continuous nonnegative increasing process L, starting from zero, with the following properties. For each fixed s,*

$$L_s = |B_s| - \int_0^s \operatorname{sgn}(B_u)\, dB_u \quad \text{in } L^2. \tag{1}$$

Its Stieltjes measure dL is concentrated on $\{s : B_s = 0\}$. For every $\kappa \geq 0$, set

$$A_\kappa(s) = s + \kappa L_s, \qquad C_\kappa = A_\kappa^{-1}, \qquad X_t^\kappa = B_{C_\kappa(t)}. \tag{2}$$

The clock and process have measurable continuous path realizations. For almost every sample, simultaneously for all $\kappa, t \geq 0$,

$$H_\kappa(t) := \int_0^t \mathbf{1}_{\{X_r^\kappa = 0\}}\, dr = \kappa L_{C_\kappa(t)} = t - C_\kappa(t). \tag{3}$$

For each fixed κ, write

$$\mathcal{G}_t^0 = \sigma(X_r^\kappa : 0 \leq r \leq t), \qquad \mathcal{G}_t = \mathcal{G}_t^0 \vee \mathcal{N}.$$

The operational clock $C_\kappa(t)$ is $\mathcal{G}_t$-measurable. The process X^κ is a martingale in both $\mathcal{G}^0$ and $\mathcal{G}$, and $(X_t^\kappa)^2 - C_\kappa(t)$ is a martingale in $\mathcal{G}$. For $\kappa, t > 0$,

$$\mathbb{E}H_\kappa(t) > 0, \qquad \mathbb{E}(X_t^\kappa)^2 + \mathbb{E}H_\kappa(t) = t, \qquad \mathbb{E}(X_t^\kappa)^2 < t. \tag{4}$$

In particular, $\mathbb{P}(H_\kappa(t) > 0) > 0$.

The last assertion is positive probability, rather than an assertion that holding is positive on every sample or that a terminal atom has been computed at each time. At $\kappa = 0$, $C_0(t) = t$ and $X^0 = B$. All clocks in the theorem use the same Brownian sample. Independence of the clock and the Brownian motion is neither true nor needed.

The time-change construction is classical. The appendix of Ayala–Carinci–Redig [1], equations (112)–(114), explicitly uses Brownian time plus a local-time term and its inverse to construct two-sided sticky Brownian motion. Bass [2], Section 2, equations (2.3)–(2.4), places inverse additive clocks in the classical speed-measure construction. Our contribution is an exposition of the verified construction, its versions and observed-filtration statements; we make no independent priority claim for these mechanisms.

2 Smooth curvature masses and their fixed-time limit

For $e > 0$ let

$$r_e(x) = \sqrt{x^2 + e^2}, \qquad a_e(x) = \frac{x}{\sqrt{x^2 + e^2}}, \qquad b_e(x) = \frac{e^2}{(x^2 + e^2)^{3/2}}.$$

Direct differentiation gives $r_e' = a_e$, $r_e'' = b_e$. Moreover,

$$0 \leq r_e(x) - |x| \leq e, \quad |a_e(x)| \leq 1, \quad 0 \leq b_e(x) \leq e^{-1}. \tag{5}$$

Indeed, $|x| \leq r_e(x) \leq |x| + e$ follows by squaring nonnegative quantities. The slope bound follows from $|x| \leq r_e$; the curvature bound follows from $r_e \geq e$. Fix the positive sequence $e_n \downarrow 0$ used in the source, and put

$$L_t^n = \frac{1}{2}\int_0^t b_{e_n}(B_u)\, du, \qquad M_t^n = r_{e_n}(B_t) - e_n - L_t^n. \tag{6}$$

The first process is continuous, increasing and nonnegative on every continuous Brownian path. The second is continuous and, by the smooth Itô formula,

$$M_t^n = \int_0^t a_{e_n}(B_u)\, dB_u \quad \text{at each fixed time, almost surely.} \tag{7}$$

The coefficients in this formula are bounded and predictable. The curvature term is integrable on every bounded horizon by (5); thus the formula involves no limiting local time as an input.

Define $\operatorname{sgn}(0) = 0$ and let $M_t = \int_0^t \operatorname{sgn}(B_u)\, dB_u$, initially as an L^2 class for each t. The coefficient is an actual predictable field: it is the bounded pointwise limit of the predictable continuous functions $a_{e_n}(B_u)$. At every x, $a_{e_n}(x) \to \operatorname{sgn}(x)$, and $|a_{e_n}| \le 1$. Dominated convergence on $[0,T] \times \Omega$ and the stochastic isometry give

$$\int_0^T \mathbb{E}|a_{e_n}(B_u) - \operatorname{sgn}(B_u)|^2 du \to 0, \qquad M_T^n \longrightarrow M_T \quad \text{in } L^2. \tag{8}$$

Since $|r_{e_n}(B_T) - e_n - |B_T|| \le e_n$,

$$L_T^n \longrightarrow \ell_T := |B_T| - M_T \quad \text{in } L^2. \tag{9}$$

The closed positive cone in L^2 shows $\ell_T \ge 0$ almost surely. Similarly, $\ell_s \le \ell_t$ as L^2 classes whenever $s \le t$. These fixed-time statements do not yet construct a continuous increasing random path. That distinction motivates the next step.

3 A continuous increasing mass from a summable subsequence

We first give the maximal estimate needed to keep the limiting mass on the original sample space.

Lemma 3.1. *For n, m, $T \ge 0$ and $\varepsilon > 0$,*

$$\varepsilon \mathbb{P}\left(\sup_{u \le T} |M_u^n - M_u^m| > \varepsilon \right) \le \left(\int_0^T \mathbb{E}|a_{e_n}(B_u) - a_{e_m}(B_u)|^2 du \right)^{1/2}. \tag{10}$$

Consequently, L^n is Cauchy in probability uniformly on each compact time interval.

Proof. For a finite ordered set of times including T, $D = M^n - M^m$ is a martingale. Partition the first-crossing event $\max |D| > \varepsilon$ according to its first crossing time u. Each piece belongs to $\mathcal{F}_u$, and $|D_u| \le \mathbb{E}(|D_T| \mid \mathcal{F}_u)$. Summing the integrals over the disjoint pieces gives $\varepsilon \mathbb{P}(\max |D| > \varepsilon) \le \mathbb{E}|D_T|$. An increasing collection of finite sets dense in $[0,T]$ gives the continuous-time estimate by continuity and monotone convergence of events. The stochastic isometry and Cauchy–Schwarz give (10). This argument also proves measurability of the crossing event, using only countably many evaluations.

From (6) and the normalized absolute-value bound,

$$\sup_{u \le T} |L_u^n - L_u^m| \le e_n + e_m + \sup_{u \le T} |M_u^n - M_u^m|. \tag{11}$$

The right side tends to zero in probability by (8) and (10). This proves the claim. □

For each $j \ge 0$, choose N_j such that for $n, m \ge N_j$,

$$\mathbb{P}\left(\sup_{u \le j+1} |L_u^n - L_u^m| > 2^{-j} \right) < 2^{-j}.$$

Set $n_0 = N_0$ and $n_{j+1} = \max\{n_j + 1, N_{j+1}\}$. The consecutive bad events have summable probabilities, so Borel–Cantelli implies that, almost surely, for all sufficiently large j,

$$\sup_{u \le j+1} |L_u^{n_{j+1}} - L_u^{n_j}| \le 2^{-j}. \tag{12}$$

Fix any compact horizon T. For $k > j$ sufficiently large that $j + 1 \geq T$, telescoping bounds the uniform difference by $\sum_{i=j}^{k-1} 2^{-i}$. Completeness of $C([0,T])$ gives a uniform limit. These limits agree on overlapping compact intervals, giving one continuous path L on $[0,\infty)$. It is increasing, nonnegative and zero at the origin, because every approximant has these properties. The same probability-one event works for all compact horizons, by taking integer horizons first.

The event in (12) is measurable, since uniform crossings of continuous paths are countable evaluation events. Each map $\omega \mapsto L^{n_j}(\omega)$ is measurable into the continuous path space with its compact-convergence topology: evaluations are measurable, and on this separable path space they generate its Borel sigma field. Set the limiting path equal to zero on the exceptional event. The resulting random continuous path is measurable, since it is the path-space limit on a measurable event of convergence. This normalization gives a continuous increasing mass starting at zero on every sample. The source instead centers its Stieltjes cumulative function on exceptional samples; both conventions agree almost surely.

At each fixed T, almost-sure subsequence convergence to L_T and (9) identify

$$L_T = \ell_T \quad \text{almost surely}, \qquad L_T \in L^2. \tag{13}$$

For example, take a further subsequence along which the L^2 convergence is almost sure, and use uniqueness of real limits. Thus (1) holds. Define the continuous process $\widehat{M}_t = |B_t| - L_t$. At each fixed time it is a version of M_t; conditional-expectation identities transfer to this version. It is adapted to $\mathcal{F}$ and a martingale there. The identity

$$|B_t| = \widehat{M}_t + L_t \tag{14}$$

is now a simultaneous pathwise identity, rather than a substitution of a random time into unrelated fixed-time null-set statements.

4 Support of the Stieltjes mass

For a compact interval $[s,t]$ on which $|B_u| \geq \delta > 0$,

$$0 \leq L_t^n - L_s^n = \frac{1}{2}\int_s^t \frac{e_n^2}{(B_u^2 + e_n^2)^{3/2}} du \leq \frac{(t-s)e_n^2}{2\delta^3}.$$

Uniform subsequence convergence therefore gives $L_t = L_s$. Every compact interval free of Brownian zeros has a positive such δ, by continuity and compactness. Consequently the Stieltjes measure dL vanishes locally on the open set $\{B \neq 0\}$. Second countability supplies a countable cover by such neighborhoods, so

$$dL(\{s : B_s \neq 0\}) = 0. \tag{15}$$

Continuity of L implies that dL has no atoms, including at zero; its interval masses are $dL((s,t]) = L_t - L_s$, and $dL([0,t]) = L_t$.

Ordinary Brownian time gives no mass to the same zero set. Indeed, $\mathbb{P}(B_s = 0) = 0$ for every $s > 0$, since B_s has a nondegenerate Gaussian law. Tonelli yields

$$\mathbb{E}\int_0^N \mathbf{1}_{\{B_s=0\}} ds = \int_0^N \mathbb{P}(B_s = 0) ds = 0.$$

The integrand is nonnegative. Taking all integer N gives, on one probability-one event,

$$ds(\{s : B_s = 0\}) = 0. \tag{16}$$

The two different measures in (15) and (16) are exactly what creates physical holding.

Nontriviality follows without a distributional formula for local time. The stochastic integral is centered, so

$$\mathbb{E}L_T = \mathbb{E}|B_T| > 0 \qquad (T > 0). \tag{17}$$

Both terms are integrable by (13). Positivity uses only the nondegenerate Brownian marginal, which makes $|B_T| > 0$ almost surely. We use no assertion that L_T is positive on every sample.

The spatial image of this temporal mass is quite different. On the same probability-one event as (15), define, for every $T \geq 0$, the finite Borel measure

$$\rho_T = B_*(dL|_{[0,T]}).$$

For each Borel set $E \subset \mathbb{R}$, concentration on the Brownian zero set gives

$$\rho_T(E) = \int_{[0,T]} \mathbf{1}_E(B_s)\, dL_s = \mathbf{1}_E(0) L_T, \qquad \rho_T = L_T \delta_0. \tag{18}$$

This is a pathwise measure identity, simultaneously for all T; it needs no separate exceptional event for each E. Indeed the map $s \mapsto B_s$ equals the constant zero map dL-almost everywhere, so their image measures agree on every Borel set. An atomless measure in time can thus have a spatial atom: the observation is constant on the set carrying its mass. In particular $\rho_T((1,2]) = 0$, so ρ_T is never spatial Lebesgue measure. For each fixed $T > 0$, (17) also gives $\mathbb{P}(L_T > 0) > 0$; on that event ρ_T is not absolutely continuous with respect to spatial Lebesgue measure. No assertion that this atom is nonzero on every sample is needed.

5 Invertible clocks and exact measure transport

We record the deterministic clock mechanism in a form that can be checked before inserting any randomness.

Lemma 5.1. *If $l : [0,\infty) \to [0,\infty)$ is continuous, increasing and $l(0) = 0$, then $A(s) = s + \kappa l(s)$ is a continuous strictly increasing bijection of $[0,\infty)$ onto itself. Its inverse C is continuous, increasing, and satisfies*

$$C(t) \leq t, \qquad 0 \leq C(t) - C(r) \leq t - r \quad (r \leq t), \qquad C(t) + \kappa l(C(t)) = t. \tag{19}$$

Proof. For $v > u$, $A(v) - A(u) \geq v - u > 0$. Also $A(0) = 0$ and $A(s) \geq s \to \infty$; the intermediate value theorem gives surjectivity. The inverse of a continuous increasing bijection of an interval is continuous and increasing. Substitute $C(t)$ into A to get the balance identity and $C(t) \leq t$. Subtract the balance identities at r, t; the nonnegative mass increment gives the upper increment bound. The lower bound follows from monotonicity. □

Apply this to each mass sample. For fixed κ, t, measurability of the inverse follows from

$$\{C_\kappa(t) < b\} = \{t < A_\kappa(b)\}, \qquad b \geq 0. \tag{20}$$

Each right side is measurable. Continuous time paths and measurable coordinates give jointly measurable clocks and measurable continuous path maps. Evaluation $(p, s) \mapsto p(s)$ is continuous for compact convergence on bounded time neighborhoods. Hence $X_t^\kappa = B_{C_\kappa(t)}$ has measurable coordinates, jointly measurable time/sample evaluation, and a measurable continuous path realization. No random time is selected using an independent copy of B.

On $[0,\infty)$ let ds denote Lebesgue measure with no atom at zero. For any continuous mass path from the construction, define $\eta_\kappa = ds + \kappa\, dL$. Then

$$(A_\kappa)_\# \eta_\kappa = dt. \tag{21}$$

To prove this, test $[0,t]$. Strict invertibility gives $A_\kappa^{-1}([0,t]) = [0, C_\kappa(t)]$, whose η_κ-mass is $C_\kappa(t) + \kappa L_{C_\kappa(t)} = t$. These intervals form a generating pi system and exhaust the half-line by finite-mass integer intervals. Uniqueness of locally finite measures proves (21) on every Borel set. This is a measure identity, so it also applies to each sample's own Borel zero set.

The complete spatial occupation measure

The same pathwise transport determines more than time spent at zero. For a fixed good sample define finite Borel measures

$$\mathcal{O}_T^X = (X^\kappa)_\#(dt|_{[0,T]}), \qquad \mathcal{O}_S^B = B_\#(ds|_{[0,S]}).$$

Then, simultaneously for all $\kappa, T \geq 0$,

$$\mathcal{O}_T^X = \mathcal{O}_{C_\kappa(T)}^B + \kappa L_{C_\kappa(T)} \delta_0. \tag{22}$$

Indeed restriction of (21) gives $dt|_{[0,T]} = (A_\kappa)_\#((ds + \kappa dL)|_{[0,C_\kappa(T)]})$. Push this identity forward by X^κ, use $X^\kappa \circ A_\kappa = B$, distribute the image measure over the two finite measures, and apply (18) to the second one. All these are equalities of measures on the same good path; thus no test-dependent exceptional sets are introduced.

`BrownianTimeChangedOccupationMeasure.occupation_decomposition`.

Consequently every bounded continuous $g : \mathbb{R} \to \mathbb{R}$ satisfies

$$\int_0^T g(X_t^\kappa)\, dt = \int_0^{C_\kappa(T)} g(B_s)\, ds + \kappa L_{C_\kappa(T)} g(0). \tag{23}$$

Both ordinary integrals exist by continuity on compact intervals; the measure identity also proves this for every bounded Borel g. The latter assertion is a human consequence of (22); the named integral theorem uses bounded continuous tests. Nonnegative Borel tests are allowed as extended integrals as well. No integrability of an arbitrary unbounded signed test is inferred. Taking total masses gives $T = C_\kappa(T) + \kappa L_{C_\kappa(T)}$, while testing the singleton zero gives the holding identity below, since the Brownian ordinary occupation of zero vanishes. Thus a temporally singular clock measure becomes an explicit spatial atom in the changed occupation measure.

`BrownianOccupationIntegralTransport.bounded_observation_transport`.

6 Holding time and a strict second-moment budget

Fix a sample on which (15) and (16) hold. For any $\kappa, t \geq 0$, let $Z = \{s : B_s = 0\}$ and $E_t = \{r \leq t : X_r^\kappa = 0\}$. These sets are Borel, and

$$A_\kappa^{-1}(E_t) = Z \cap [0, C_\kappa(t)].$$

By (21),

$$\begin{aligned} \int_0^t \mathbf{1}_{\{X_r^\kappa = 0\}} dr &= \eta_\kappa(Z \cap [0, C_\kappa(t)]) \\ &= \kappa\, dL([0, C_\kappa(t)]) = \kappa L_{C_\kappa(t)} = t - C_\kappa(t). \end{aligned}$$

The ordinary-time contribution is zero by (16); concentration of dL on Z gives the second line. The proof uses just two fixed probability-one path properties. Thus it holds simultaneously for all uncountably many κ, t without an uncountable intersection of null sets.

For $\kappa > 0$ and each sample, strict inversion also gives

$$t - C_\kappa(t) > 0 \iff C_\kappa(t) < t \iff t < A_\kappa(t) \iff L_t > 0. \tag{24}$$

For $t > 0$, (17) implies $\mathbb{P}(L_t > 0) > 0$. Since $0 \le H_\kappa(t) \le t$, it is integrable; (24) implies $\mathbb{E}H_\kappa(t) > 0$. The second-moment part of (4) will follow from bounded sampling below. This separates its probabilistic input from the pathwise measure calculation.

7 Bounded sampling with the clock's own Brownian noise

The continuous mass L_s agrees at each fixed time with the raw Brownian-past L^2 class in (1). Hence L_s is $\mathcal{F}_s$-measurable under our ambient null augmentation. The same is true of $A_\kappa(s)$. The inverse is a bounded stopping time because

$$\{C_\kappa(t) \le s\} = \{t \le A_\kappa(s)\} \in \mathcal{F}_s, \qquad C_\kappa(t) \le t. \tag{25}$$

The weak inequality here, as opposed to the strict inequality in (20), is required for the stopping-time sigma field.

We give the bounded sampling argument used in the construction. If Y is a continuous martingale and $\sigma \le \tau \le T$ are stopping times, with integrable compact-time maximum, then

$$\mathbb{E}(Y_\tau \mid \mathcal{F}_\sigma) = Y_\sigma. \tag{26}$$

Round both times up to a dyadic grid, obtaining finite-valued stopping times $\sigma_n \le \tau_n \le T+1$. For a finite grid, (26) follows by writing the sampled difference as a sum of martingale increments multiplied by past-measurable indicators. For any $E \in \mathcal{F}_\sigma$ we also have $E \in \mathcal{F}_{\sigma_n}$, so $\mathbb{E}\mathbf{1}_E Y_{\tau_n} = \mathbb{E}\mathbf{1}_E Y_{\sigma_n}$. Path continuity gives almost-sure convergence of both values. Their absolute values are bounded by $\sup_{u \le T+1} |Y_u|$, so dominated convergence passes this equality to the limit. Stopped values are $\mathcal{F}_\sigma$-measurable: a continuous adapted process is progressively measurable, and its evaluation at a stopping time is measurable in the stopping-time sigma field. This last assertion follows by approximating the continuous path on each deterministic interval by time-step functions, then restricting to $\{\sigma \le s\}$. Integrability and the integral identity for every E prove (26).

The maximum condition holds for the processes used here. The L^2 martingales B and $\widehat{M}$ have integrable compact maxima by the finite-grid L^2 maximal inequality and path continuity. For completeness, that inequality follows by the same first-crossing partition with levels $a > 0$: $a\mathbb{P}(Y^* > a) \le \mathbb{E}(|Y_T|\mathbf{1}_{Y^* > a})$. Integrate twice the resulting inequality in a, then use Cauchy–Schwarz to get $\mathbb{E}(Y^*)^2 \le 2\mathbb{E}(|Y_T|Y^*) \le 2\|Y_T\|_2\|Y^*\|_2$, first with truncated maxima and then by monotone convergence. Thus $\|Y^*\|_2 \le 2\|Y_T\|_2$. Finally,

$$\sup_{u \le T} |B_u^2 - u| \le \sup_{u \le T} B_u^2 + T$$

is integrable. So sampling applies also to the martingale $B_s^2 - s$.

Let $\mathcal{F}_{C_\kappa(t)}$ be the stopping-time sigma field and denote this increasing physical filtration by $\mathcal{H}_t$. Applying (26) at $C_\kappa(r) \le C_\kappa(t)$ gives martingales

$$X_t^\kappa, \qquad (X_t^\kappa)^2 - C_\kappa(t), \qquad \widehat{M}_{C_\kappa(t)} \quad \text{in } \mathcal{H}. \tag{27}$$

Their sampled values are adapted by stopped-value measurability. Sampling the centered square martingale gives $\mathbb{E}(X_t^\kappa)^2 = \mathbb{E}C_\kappa(t)$; its integrability follows from the preceding compact-maximum bound. Combine this equality with (3) to obtain (4), including its strict inequality. The process is centered by the same argument applied to B, so its second moment is indeed its variance.

The pathwise Tanaka identity also becomes

$$|X_t^\kappa| = \widehat{M}_{C_\kappa(t)} + L_{C_\kappa(t)}, \qquad \kappa L_{C_\kappa(t)} = t - C_\kappa(t).$$

This uses (14) on the common path event. Substitution of the random time would not be justified from fixed-time L^2 identities alone.

8 Causal clock recovery and the observed martingales

For a continuous real path p define

$$h(p,t) = \int_0^t \mathbf{1}_{\{p(r)=0\}} dr, \qquad c(p,t) = t - h(p,t). \tag{28}$$

Then $0 \le h(p,t) \le t$ and $0 \le c(p,t) \le t$ for every path. Both functionals are jointly Borel measurable in (p,t). Indeed,

$$(p,t,r) \longmapsto \mathbf{1}_{\{0<r\le t\}} \mathbf{1}_{\{p(r)=0\}}$$

is Borel measurable, since evaluation is continuous and the indicated sets are Borel. Integrating this nonnegative function in r proves measurability by the measurable-integral theorem. The integral is finite, so subtraction in (28) is legitimate. Both functionals depend only on $p|_{[0,t]}$.

To make the raw-past measurability explicit, the map $\omega \mapsto X^\kappa(\omega)|_{[0,t]}$ into $C([0,t])$ is $\mathcal{G}_t^0$-measurable: its evaluations are precisely the coordinates generating $\mathcal{G}_t^0$, and they generate the Borel sigma field of $C([0,t])$. Extend a prefix constantly after t; this is a measurable map back into the full continuous path space. Applying (28) to the extension therefore gives a $\mathcal{G}_t^0$-measurable version of the recovered time. By (3),

$$c(X^\kappa, t) = C_\kappa(t) \quad \text{almost surely.} \tag{29}$$

The actual clock coordinate is ambient measurable. A measurable variable equal almost surely to a raw-past measurable one is measurable in $\mathcal{G}_t^0 \vee \mathcal{N}$. Thus the actual clock is adapted to $\mathcal{G}$, with a causal raw-past version. We do not silently identify these two assertions on unaugmented sigma fields.

For $u \le t$, the coordinate X_u^κ belongs to $\mathcal{H}_u \subseteq \mathcal{H}_t$, so $\mathcal{G}_t^0 \subseteq \mathcal{H}_t$. The elementary shrinking-filtration argument now transfers the first martingale in (27): for $r \le t$,

$$\mathbb{E}(X_t^\kappa \mid \mathcal{G}_r^0) = \mathbb{E}(\mathbb{E}(X_t^\kappa \mid \mathcal{H}_r) \mid \mathcal{G}_r^0) = \mathbb{E}(X_r^\kappa \mid \mathcal{G}_r^0) = X_r^\kappa.$$

It remains a martingale after ambient null augmentation, because adjoining ambient null events changes neither integrals nor conditional expectations up to almost-sure equality.

For the compensated square, use the augmented physical filtration $\mathcal{H}_t \vee \mathcal{N}$. Then $\mathcal{G}_t \subseteq \mathcal{H}_t \vee \mathcal{N}$, and $(X_t^\kappa)^2 - C_\kappa(t)$ is $\mathcal{G}_t$-measurable by (29). The same tower argument transfers the second martingale in (27) to $\mathcal{G}$. This completes every assertion of Theorem 1.1.

9 A balanced coupling of two genuine Brownian coordinates

The same inverse clock has a useful consequence beyond the single sticky path. On the product of two independent Brownian source spaces, let W be independent of the Brownian motion B used to construct L and C_κ. For the rest of this section fix κ and abbreviate

$$C(t) = C_\kappa(t), \qquad D(t) = 2t - C(t), \qquad S_t = B_{C(t)}, \quad V_t = W_{D(t)}.$$

The inverse increment bound in (19) implies that D is continuous and increasing, with $t \le D(t) \le 2t$, $D(0) = 0$, and $C(t) + D(t) = 2t$. Define

$$Y_t^+ = \frac{V_t + S_t}{\sqrt{2}}, \qquad Y_t^- = \frac{V_t - S_t}{\sqrt{2}}. \tag{30}$$

These are measurable continuous paths, by measurable evaluation and the product probability law. The source centers both Brownian paths at their initial values on exceptional samples; they start at zero almost surely, so this convention gives the same path law.

Theorem 9.1. *Each coordinate Y^+ and Y^- has the law of standard Brownian motion, including all finite collections of independent Gaussian increments. For every $t \ge 0$,*

$$\operatorname{Cov}(Y_t^+, Y_t^-) = \mathbb{E}H_\kappa(t) = \mathbb{E}\int_0^t \mathbf{1}_{\{Y_r^+ = Y_r^-\}} dr. \tag{31}$$

For $\kappa, t > 0$ the covariance is strictly positive, so the terminal coordinates are not independent. For every $\kappa, T > 0$ there exists a deterministic $s \in (0, T]$ for which

$$\mathbb{P}(Y_s^+ = Y_s^-) > 0. \tag{32}$$

At such s, the joint law differs from the product of its Gaussian marginals, which gives zero mass to the diagonal.

The assertion that each coordinate is Brownian concerns its own full path law. We do not deduce a Brownian property relative to the joint pair filtration merely from the marginal assertion. Nor do we assert a stochastic flow from this single two-point construction.

Why a single Gaussian marginal does not prove the theorem

Given the entire B sample, the clock $D(t)$ is deterministic and $W_{D(t)}$ is conditionally Gaussian. But $B_{C(t)}$ is sampled at a stopping time depending on B itself. Treating it as an independent Gaussian with variance $C(t)$ is invalid. The compensating Fourier martingale, rather than that invalid conditional independence, gives the cancellation in the proof below.

For $v \in \mathbb{R}$ define the complex process

$$Z_v(s) = \exp\bigl(v^2 s/2 + ivB_s\bigr). \tag{33}$$

This is a continuous complex martingale on every bounded time interval. Indeed, Brownian independent increments give

$$\mathbb{E}(e^{iv(B_t - B_s)} \mid \mathcal{F}_s) = e^{-v^2(t-s)/2},$$

which verifies the martingale identity directly. Its modulus is the deterministic quantity $e^{v^2 s/2}$; its real and imaginary parts therefore satisfy bounded sampling as in Section 7. It is never zero and $|Z_v(s)| \ge 1$.

Lemma 9.2. *For ordered bounded Brownian stopping times $\tau_0 \le \cdots \le \tau_n \le T$ and real $v_0, \dots, v_{n-1}$,*

$$\mathbb{E} \prod_{j=0}^{n-1} \frac{Z_{v_j}(\tau_{j+1})}{Z_{v_j}(\tau_j)} = 1. \tag{34}$$

Proof. Write the product through $j = k-1$ as R_k, with $R_0 = 1$. It is $\mathcal{F}_{\tau_k}$-measurable. Every factor has modulus

$$\exp\bigl(v_j^2(\tau_{j+1} - \tau_j)/2\bigr) \le e^{v_j^2 T/2},$$

so the finite product is bounded by a deterministic constant. Also $1/Z_{v_k}(\tau_k)$ is bounded and $\mathcal{F}_{\tau_k}$-measurable. Bounded sampling of Z_{v_k}, separately in real and imaginary parts, gives

$$\begin{aligned} \mathbb{E}R_{k+1} &= \mathbb{E}\left[\frac{R_k}{Z_{v_k}(\tau_k)} \mathbb{E}(Z_{v_k}(\tau_{k+1}) \mid \mathcal{F}_{\tau_k})\right] \\ &= \mathbb{E}R_k. \end{aligned}$$

All products are integrable by the deterministic bounds. Induction proves (34); independence of the stopping intervals has not been used. □

10 The full finite-increment law by exact compensation

More generally let $a, b \in \mathbb{R}$ satisfy $a^2 = b^2$, and put $Q_t = aV_t + bS_t$. For deterministic ordered times $0 \le t_0 \le \cdots \le t_n$ and real $u_0, \dots, u_{n-1}$, we prove

$$\mathbb{E} \exp\left(i \sum_{j=0}^{n-1} u_j (Q_{t_{j+1}} - Q_{t_j}) \right) = \prod_{j=0}^{n-1} e^{-(u_j a)^2 (t_{j+1} - t_j)}. \tag{35}$$

Condition first on the full B sample. The $D(t_j)$ are then ordered deterministic times for the independent Brownian motion W. Its Gaussian independent increments give the conditional expectation

$$\prod_{j=0}^{n-1} e^{-(u_j a)^2 (D(t_{j+1}) - D(t_j))/2} e^{i u_j b (B_{C(t_{j+1})} - B_{C(t_j)})}. \tag{36}$$

This use of conditioning can equally be written as an iterated integral on the product of the two source spaces. The original integrand has modulus one, so Fubini is valid.

Write $\tau_j = C(t_j)$, $\Delta t_j = t_{j+1} - t_j$ and $\Delta\tau_j = \tau_{j+1} - \tau_j$. Since $\Delta D_j = 2\Delta t_j - \Delta\tau_j$ and $(u_j a)^2 = (u_j b)^2$, each factor in (36) is exactly

$$e^{-(u_j a)^2 \Delta t_j} \exp\Bigl((u_j b)^2 \Delta\tau_j/2 + i u_j b (B_{\tau_{j+1}} - B_{\tau_j}) \Bigr) = e^{-(u_j a)^2 \Delta t_j} \frac{Z_{u_j b}(\tau_{j+1})}{Z_{u_j b}(\tau_j)}. \tag{37}$$

The times τ_j are genuine ordered stopping times, all bounded by t_n. Lemma 9.2 shows that the expectation of the product of the phase ratios is one. Pulling out the deterministic factors proves (35).

For $a = 1/\sqrt{2}$ and $b = \pm 1/\sqrt{2}$, the right side becomes

$$\prod_{j=0}^{n-1} \exp(-u_j^2 \Delta t_j/2),$$

the characteristic function of a vector of independent centered Gaussian increments with variances Δt_j. Equality of characteristic functions determines finite-dimensional probability laws. Thus the increment vectors of Y^+ and Y^- separately have exactly those Brownian laws, including zero-length intervals. Both paths are continuous and start at zero. This proves the full Brownian-coordinate assertion in Theorem 9.1, without inferring it from a one-time Gaussian calculation.

The proof remains valid when $t_0 > 0$: the first phase ratio is conditioned at $C(t_0)$, and the empty product at that time is one. All parameter-dependent bounds are finite for the fixed collection of frequencies and times. No limiting product, random signed law or independence of the Brownian stopping-clock increments is used.

11 Covariance, coincidence and the deterministic-time diagonal

The elementary pathwise identity

$$Y_t^+ - Y_t^- = \sqrt{2} S_t \tag{38}$$

implies

$$\{Y_t^+ = Y_t^-\} = \{S_t = 0\}, \qquad \int_0^t \mathbf{1}_{\{Y_r^+ = Y_r^-\}} dr = H_\kappa(t)$$

on the same common event as (3). The product-space extension preserves that event because the second source is an independent probability factor. Each Brownian coordinate is centered with second moment t. From (38) and the exact single-process variance,

$$\mathbb{E}(Y_t^+ - Y_t^-)^2 = 2\mathbb{E}S_t^2 = 2\mathbb{E}C(t).$$

Expand the square. The product $Y_t^+ Y_t^-$ is integrable by Cauchy–Schwarz, so the calculation is valid and gives

$$2t - 2\mathbb{E}(Y_t^+ Y_t^-) = 2\mathbb{E}C(t), \qquad \mathrm{Cov}(Y_t^+, Y_t^-) = t - \mathbb{E}C(t) = \mathbb{E}H_\kappa(t).$$

This proves (31) and, for $\kappa, t > 0$, strict positivity. Independence would force zero covariance for these L^2 variables, so it is impossible at every such terminal time.

To prove the stronger diagonal statement at some deterministic time in each horizon, Tonelli gives

$$\int_0^T \mathbb{P}(S_s = 0) ds = \mathbb{E}H_\kappa(T) > 0. \tag{39}$$

Joint measurability of S was proved in Section 5. If every deterministic $s \in (0, T]$ had zero boundary probability, the integral would vanish. Hence some such s satisfies $\mathbb{P}(S_s = 0) > 0$, and (38) gives (32). The product of two nondegenerate Gaussian marginals has zero diagonal mass: integrate the singleton mass $\mathbb{P}(Y_s^- = x) = 0$ over the first Gaussian marginal. Thus the joint law cannot be their product. This completes Theorem 9.1.

The covariance conclusion holds at every positive deterministic time, whereas the diagonal conclusion just proved is existential within each positive horizon. The two quantifiers should not be interchanged. Classical sticky Brownian couplings already combine Brownian coordinates with coincidence occupation. Howitt–Warren [3], equations (1)–(2), formulate such couplings using a common-filtration bracket and local time. The present source-mapped claims establish the stated coordinate path laws, covariance and coincidence facts; no identification of a joint bracket or of an entire stochastic flow is silently added.

12 Operational Hermite martingales and boundary tests

Fix $\kappa \geq 0$ and suppress it from C and X. The preceding clock recovery makes $C(t)$ measurable in $\mathcal{G}_t$. This permits polynomial time-space tests with their actual random time argument. Define $H_n(x,s)$ by

$$e^{zx-sz^2/2} = \sum_{n=0}^{\infty} H_n(x,s)\frac{z^n}{n!}.$$

Thus $H_0 = 1$, $H_1 = x$, and differentiation in z gives $H_{n+1} = xH_n - nsH_{n-1}$.

Proposition 12.1. *For every integer $n \geq 0$, the continuous process $H_n(X_t, C(t))$ is a martingale in the augmented observed past $\mathcal{G}$. In particular,*

$$X_t^4 - 6C(t)X_t^2 + 3C(t)^2$$

is integrable and is such a martingale.

Proof. For deterministic $0 \leq s \leq t$, independence and the Gaussian moment generating function of $B_t - B_s$ imply

$$\mathbb{E}[e^{zB_t - tz^2/2} \mid \mathcal{F}_s] = e^{zB_s - sz^2/2}.$$

Differentiating n times at zero gives the conditional identity for $H_n(B_t, t)$. This differentiation can be justified after pairing with any bounded $\mathcal{F}_s$-measurable variable: derivatives on a fixed bounded real z interval are bounded by a polynomial in $|B_t|$ times $e^{a|B_t|+a^2t/2}$ for a finite a. This is integrable by the Gaussian density. Dominated convergence therefore justifies differentiation and characterizes the conditional expectation.

On any fixed horizon $[0,T]$, this continuous polynomial martingale has a square-integrable terminal value. Its maximum is integrable by the L^2 maximal inequality. The bounded-sampling argument of Section 7 applies to the stopping times $C(s) \leq C(t) \leq T$. Consequently $H_n(X_t, C(t))$ is a martingale in the physical past $\mathcal{F}_{C(t)}$. It is adapted to $\mathcal{G}$, since X_t and $C(t)$ are $\mathcal{G}_t$-measurable. For $A \in \mathcal{G}_s \subseteq \mathcal{F}_{C(s)}$ the sampled martingale integral identity against $\mathbf{1}_A$ now proves the $\mathcal{G}$ martingale identity. Ambient null augmentations do not change these integrals. Finally the recurrence gives $H_4(x,s) = x^4 - 6sx^2 + 3s^2$. □

No independence between C and B was used. The time argument cannot simply be replaced by t: for $\kappa, t > 0$,

$$\mathbb{E}(X_t^2 - t) = -\mathbb{E}H_\kappa(t) < 0,$$

so $X_t^2 - t$, starting at zero, is not a martingale.

We next obtain a true boundary martingale from a smooth test and Tanaka's formula. Let $f \in C^2(\mathbb{R})$ with bounded f' and f''. No boundedness of f is required. Set

$$a = \frac{\kappa}{2}f''(0), \qquad F_\kappa(x) = f(x) + a|x|, \qquad g(x) = \frac{1}{2}f''(x). \tag{40}$$

Proposition 12.2. *The process*

$$F_\kappa(X_t) - F_\kappa(0) - \int_0^t g(X_s)\,ds$$

is a continuous martingale in $\mathcal{G}$. Off zero, $\frac{1}{2}F_\kappa'' = g$, and its one-sided derivatives satisfy

$$F_\kappa'(0+) - F_\kappa'(0-) = 2\kappa g(0). \tag{41}$$

Proof. The original Brownian processes

$$M_u^f = f(B_u) - f(0) - \frac{1}{2}\int_0^u f''(B_v)\,dv, \qquad M_u^{\mathrm{sgn}} = |B_u| - L_u$$

are true square-integrable martingales on bounded horizons. For the first, the smooth Ito formula gives the stochastic integral of $f'(B)$; its second moment is at most $\|f'\|_\infty^2 u$. For the second, the Tanaka construction gives the integral of $\mathrm{sgn}(B)$, with second moment at most u. Both therefore admit the same bounded sampling at $C(t)$.

The exact change-of-clock measure identity proved in Section 5 is $dC(s) = \mathbf{1}_{\{X_s\neq 0\}}ds$. It gives

$$\frac{1}{2}\int_0^{C(t)} f''(B_v)\,dv = \int_0^t g(X_s)\mathbf{1}_{\{X_s\neq 0\}}\,ds.$$

The remaining physical-time drift is

$$\int_0^t g(X_s)\mathbf{1}_{\{X_s=0\}}ds = g(0)H_\kappa(t) = aL_{C(t)}.$$

This calculation is also valid at $\kappa = 0$, without division by κ. Subtracting both drift parts yields exactly

$$F_\kappa(X_t) - F_\kappa(0) - \int_0^t g(X_s)ds = M^f_{C(t)} + aM^{\mathrm{sgn}}_{C(t)}.$$

The right side is a martingale in the physical past. The left side is adapted to $\mathcal{G}$: continuity of $g(X)$ makes its time integral a limit of finite prefix Riemann sums. It is integrable since F_κ has at most linear growth and g is bounded. The same shrinking-filtration argument as above proves the claimed observed-past martingale. Finally differentiation on the two half lines gives $F'_\kappa(0\pm) = f'(0) \pm a$, and differentiating a second time off zero gives $F''_\kappa = f''$. These prove both boundary identities. □

For a concrete admissible test, take $f(x) = \sqrt{1+x^2}$. Direct calculus gives $|f'(x)| \le 1$ and $f''(x) = (1+x^2)^{-3/2} \le 1$. Thus

$$\sqrt{1+X_t^2} + \frac{\kappa}{2}|X_t| - 1 - \frac{1}{2}\int_0^t (1+X_s^2)^{-3/2}ds$$

is a true observed-past martingale. Its test has derivatives $\kappa/2$ and $-\kappa/2$ at $0+$ and $0-$, respectively, and its drift takes value $1/2$ at zero. For $\kappa > 0$ this linear-growth test is not in $C_0(\mathbb{R})$ and has a derivative jump. It is not an assertion about the domain of a closed C_0 generator.

The time-change construction and sticky derivative-jump mechanism are classical; see Ayala–Carinci–Redig [1], Appendix 6.2, equations (112)–(129). Their displayed generator convention in (121) uses the Laplacian. Here the original Brownian generator is $\frac{1}{2}d^2/dx^2$ and the local-time normalization is fixed by (1); the factors in (40)–(41) are proved directly. No generator-domain equivalence between normalizations is assumed.

Even in the nonsticky case a terminal law is not automatically a symmetry measure for the generator. When $\kappa = 0$ and $T > 0$, $X_T = B_T$ has law $\mu_T = N(0,T)$. For the bounded smooth tests $f(x) = \cos x$, $h(x) = 1$ and $L = \frac{1}{2}d^2/dx^2$,

$$\int (Lf)h\,d\mu_T = -\tfrac{1}{2}e^{-T/2} \neq 0 = \int f(Lh)\,d\mu_T.$$

The Gaussian characteristic function proves the nonzero value. This is a pairing obstruction, not a classification of invariant measures or a closed generator theorem.

13 A dense value domain with a fixed graph-norm gap

The distinction between martingale tests and a closed-generator core can be made quantitative. We work in the real Banach space $C_0(\mathbb{R})$ with its uniform norm. Fix $\kappa > 0$. Call a pair $(F, G) \in C_0(\mathbb{R})^2$ *admissible* if F is twice differentiable off zero,

$$G(x) = \tfrac{1}{2}F''(x) \quad (x \neq 0), \qquad F'(0+) - F'(0-) = 2\kappa G(0), \tag{42}$$

where both one-sided derivatives are actual finite one-sided derivatives. Their existence is a hypothesis, not a symbolic boundary convention. Continuity of G fixes its value at zero uniquely from its off-zero values. Thus the linear space of admissible pairs is the graph of a single-valued linear operator A_κ on its actual first-coordinate domain. Linearity follows by adding or scaling the off-zero identities and the two boundary derivatives.

The uniform resolvent estimate and genuine closability

An admissible F attains its positive maximum or negative minimum when its uniform norm is nonzero: vanishing at infinity restricts the relevant extremum to a compact interval. At a maximum away from zero the ordinary second-derivative test gives $G \leq 0$. At a maximum at zero its right slope is nonpositive and its left slope nonnegative; hence (42), with $\kappa > 0$, again gives $G(0) \leq 0$. Applying the same argument to $-F$ handles minima. Evaluating $qF - G$ at a point attaining $|F| = \|F\|_\infty$ therefore proves

$$q\|F\|_\infty \leq \|qF - G\|_\infty, \qquad q > 0. \tag{43}$$

In particular any two admissible solutions of the same resolvent equation have the same F: subtract their pairs and use this inequality. This is uniqueness and stability when solutions exist, not their existence.

The domain is dense. To prove this explicitly let

$$\mathcal{E} = \{g \in C_c^\infty(\mathbb{R}) : g \text{ is constant near } 0\}.$$

Every $g \in \mathcal{E}$ is admissible with $A_\kappa g = g''/2$ and $(A_\kappa g)(0) = 0$. For any $F \in C_0(\mathbb{R})$ and $\epsilon > 0$ first choose a smooth compactly supported u with $\|u - F\|_\infty < \epsilon/2$. Such a choice follows by cutting off the small tails of F and convolving the compact continuous result with a positive smooth approximate identity; uniform continuity gives uniform convergence. Continuity at zero supplies $\delta > 0$ with $|F(x) - F(0)| < \epsilon/2$ on $|x| < \delta$. Choose a smooth cutoff $0 \leq \chi \leq 1$, equal to one near zero and supported in that interval. Then

$$g = \chi F(0) + (1 - \chi)u \in \mathcal{E}, \quad |g - F| \leq \chi|F(0) - F| + (1 - \chi)|u - F| < \epsilon.$$

The support is compact since both χ and u have compact support. This proves value-norm density, including without the restriction $\kappa > 0$.

For completeness, the actual relation is closable; density plus the resolvent estimate gives a direct proof. Let Γ be its graph. The inequality (43) persists in its product-norm closure. Suppose $(0, V) \in \overline{\Gamma}$. For an original pair $(F, G) \in \Gamma$, linearity puts $(F, G + qV)$ in $\overline{\Gamma}$. Consequently

$$q\|F\|_\infty \leq \|q(F - V) - G\|_\infty \leq q\|F - V\|_\infty + \|G\|_\infty.$$

Send $q \to \infty$ with this pair fixed to get $\|F\|_\infty \leq \|F - V\|_\infty$. Now approximate V in uniform norm by first coordinates of admissible pairs, using the density just proved. The inequality forces $\|V\|_\infty = 0$. Thus the closed graph has no nonzero vertical pair, so it defines a closed linear operator $\overline{A}_\kappa$, extending A_κ and retaining a dense domain and (43). No range condition or semigroup identification has entered this argument.

An explicit admissible pair missed by all flat tests

Put

$$a = 1 + \kappa/4, \quad b = 1 + \kappa/2, \quad f_\kappa(x) = a\sqrt{1+x^2} - b\sqrt{4+x^2},$$
$$F_\kappa(x) = f_\kappa(x) + \frac{\kappa}{4}|x|, \qquad G_\kappa(x) = \frac{1}{2}\left(\frac{a}{(1+x^2)^{3/2}} - \frac{4b}{(4+x^2)^{3/2}}\right). \tag{44}$$

The smooth base has bounded first and second derivatives and $f_\kappa''(0) = a - b/2 = 1/2$. Its sticky correction is therefore exactly the $\kappa f''(0)|x|/2$ correction of Section 12. The resulting pair is admissible, and the actual observed-past martingale there applies. It has

$$G_\kappa(0) = \tfrac{1}{4}, \qquad F_\kappa'(0+) = \kappa/4, \quad F_\kappa'(0-) = -\kappa/4, \qquad F_\kappa(0) = -1 - 3\kappa/4.$$

Thus its derivative jump is $\kappa/2 = 2\kappa G_\kappa(0)$. It remains to check that both components really vanish at infinity. The second does so directly. For the first use $a - b = -\kappa/4$ to write

$$F_\kappa(x) = a(\sqrt{1+x^2} - |x|) - b(\sqrt{4+x^2} - |x|).$$

For $r > 0$ the identity $\sqrt{r^2+x^2} - |x| = r^2/(\sqrt{r^2+x^2} + |x|)$ proves that both terms tend to zero. Continuity at zero is immediate. Hence (F_κ, G_κ) is an actual nonzero pair in $C_0(\mathbb{R})^2$, not merely a linear-growth formal boundary test.

For every $g \in \mathcal{E}$, its second derivative vanishes at zero. Therefore

$$\left\| \tfrac{1}{2}g'' - G_\kappa \right\|_\infty \geq \left| \tfrac{1}{2}g''(0) - G_\kappa(0) \right| = \tfrac{1}{4}. \tag{45}$$

So $\mathcal{E}$ is not a graph core for A_κ or its closure, even though it is uniformly dense in $C_0(\mathbb{R})$ and consists of genuine martingale tests. The same fixed gap excludes graph approximation to this pair in any closed operator extending the admissible relation. The explicit gap itself also holds at $\kappa = 0$; the closability proof above and its named relation estimate retain $\kappa > 0$.

This is a classical domain distinction with a concrete frozen witness. The corresponding source mechanisms are

`SmoothFlatC0Approximation`;
`StickyFlatTestDomain`;
`StickyVanishingBoundaryTest`;
`StickyVanishingGeneratorPair`;
`StickyGeneratorRelation`;
`StickyGeneratorLinearGraph`;
`StickyGeneratorClosure`.

The density statement is a value approximation, and the closed operator construction supplies no inverse on arbitrary C_0 right-hand sides. In particular it does not identify the generator of an all-start transition semigroup or prove uniqueness of a martingale problem. Those are precisely the separate obligations preserved below.

14 Whole-path recovery and exact Hilbert-space transport

Fix $\kappa \geq 0$ throughout this section. Put $\mathcal{P} = C([0,\infty), \mathbb{R})$ with its compact-open Polish topology, let W be the law of B on $\mathcal{P}$, and let Q_κ be the law of the actual X^κ constructed above. There are Borel maps $S_\kappa, R_\kappa : \mathcal{P} \to \mathcal{P}$ such that

$$(S_\kappa)_*\mathsf{W} = \mathsf{Q}_\kappa, \quad (R_\kappa)_*\mathsf{Q}_\kappa = \mathsf{W}, \quad R_\kappa S_\kappa = \mathrm{id} \ \ \mathsf{W}\text{-a.e.}, \quad S_\kappa R_\kappa = \mathrm{id} \ \ \mathsf{Q}_\kappa\text{-a.e.} \tag{46}$$

This assertion concerns complete trajectories, rather than equality of measures on the same path coordinates or equality of physical-time filtrations. We give the measurable construction and its consequences.

A total Borel construction and an almost-sure inverse

The smooth mass approximations from Section 2 depend only on the Brownian path: they are integrals of continuous functions of $p(s)$. Their maps into $\mathcal{P}$ are Borel, because evaluation at each rational terminal time is Borel and these evaluations generate the path-space Borel sigma field. Choose the fixed subsequence already used to construct the continuous mass. In a complete metric for compact-open convergence, the set on which the corresponding sequence of paths is Cauchy is Borel: write the Cauchy condition using only countably many metric inequalities. Define $F(p)$ as its limit there and the zero path elsewhere. Pointwise limits of measurable maps on a measurable set are measurable, so F is Borel and $F(B) = L$ almost surely as *paths*, not only at separately fixed times.

To define a clock on every path, replace $F(p)$ by zero whenever it is not nondecreasing, and call the resulting continuous path $\overline{F}(p)$. The test for being nondecreasing is Borel, since it suffices to compare rational times. Set

$$\ell_p(s) = \overline{F}(p)(s) - \overline{F}(p)(0), \quad a_p(s) = s + \kappa\ell_p(s), \quad c_p = a_p^{-1}, \quad S_\kappa(p)(t) = p(c_p(t)).$$

For every p, ℓ_p is continuous, nonnegative, nondecreasing and zero at zero. Thus a_p is a continuous strictly increasing map onto $[0, \infty)$ and has a continuous inverse. For each $t, b \geq 0$,

$$\{p : c_p(t) < b\} = \{p : t < b + \kappa\ell_p(b)\}.$$

This proves coordinate measurability of $c_p(t)$. Continuous path evaluation then proves that S_κ is Borel into $\mathcal{P}$. On the full-measure event where $F(B) = L$, this construction agrees with the original clock, hence

$$S_\kappa(B) = X^\kappa \quad \text{almost surely as paths.} \tag{47}$$

The formal totalization uses the same monotonicity replacement and nonnegative clock; choices of the raw version of F outside its agreement event have no effect on this equality.

For the inverse, the earlier occupation identity recovers $C_\kappa(t)$ measurably from the full observed path. Its inverse $A_\kappa(s)$ is likewise measurable: the event $\{A_\kappa(s) < b\}$ is $\{s < C_\kappa(b)\}$. Consequently every $B_s = X^\kappa_{A_\kappa(s)}$ is measurable with respect to the ambient-null augmentation of $\sigma(X^\kappa)$. Rational evaluations and continuity give the same statement for the entire path B.

Here is the version step that turns this augmented measurability into a Borel path map. A strongly measurable function with values in a complete separable metric space has a sequence of measurable simple approximants converging to it. Each event in the ambient-null augmentation of $\sigma(X^\kappa)$ agrees almost surely with an event in $\sigma(X^\kappa)$. Replace the finitely many defining events in every simple approximant by these raw events. There is one null set outside which all countably many replacements agree. The limit of the replaced approximants, totalized by a fixed value where necessary, is a raw $\sigma(X^\kappa)$-measurable version of B. Each such simple approximant factors through X^κ: its defining events are preimages of Borel path sets, and overlaps can be made disjoint in a fixed finite order. Take their pointwise limit on the Borel convergence set in $\mathcal{P}$, and zero elsewhere. Completeness gives a Borel map R_κ with

$$R_\kappa(X^\kappa) = B \quad \text{almost surely as paths.} \tag{48}$$

This argument uses countably many approximants, not an uncountable intersection of coordinatewise null events.

Equations (47)–(48) immediately give both pushforward laws in (46). Composing them on their common agreement event gives $R_\kappa(S_\kappa(B)) = B$ and $S_\kappa(R_\kappa(X^\kappa)) = X^\kappa$. Pushing these equalities to their respective path laws proves both almost-sure inverse identities. No assertion that the maps are inverses at every exceptional path is needed.

This recovery principle is already explicit in Bass [2], the opening paragraphs of the proof of Theorem 5.2 (preprint printed page 16): the operational Brownian path and the time-changed path are related by Borel maps in both directions. Our proof records the actual version and totalization used in the frozen construction. We do not use the remainder of that theorem's weak-uniqueness argument to identify the present physical-time martingale problem. No new path-recovery principle is claimed.

Pullback isometry and the full transported adjoint domain

Define real Hilbert spaces $H_W = L^2(\mathsf{W};\mathbb{R})$ and $H_Q = L^2(\mathsf{Q}_\kappa;\mathbb{R})$. The maps

$$U : H_Q \longrightarrow H_W, \quad Uf = f \circ S_\kappa, \qquad V : H_W \longrightarrow H_Q, \quad Vg = g \circ R_\kappa$$

are well defined on almost-everywhere equivalence classes. Indeed each pushforward identity preserves null sets and gives equality of the two squared integrals. Hence they are linear isometries; the inverse identities in (46) give $UV = I$ and $VU = I$. In particular U is onto, rather than just an isometric embedding. The same pushforwards preserve means.

Let E be a real Hilbert space and let $T : D(T) \subset H_W \to E$ be a densely defined linear operator. Its pullback has the *whole* inverse-image domain

$$D(T_Q) = U^{-1}D(T), \qquad T_Q f = T(Uf).$$

Density is preserved in both directions because U is a homeomorphism. Its graph is the inverse image of the graph of T under the homeomorphism $(f, e) \mapsto (Uf, e)$; thus closedness is also equivalent in both directions. More precisely, for the Hilbert adjoints,

$$D(T_Q^*) = D(T^*) \subset E, \qquad T_Q^* e = U^{-1}T^* e. \tag{49}$$

To prove both domain inclusions, recall that $e \in D(T^*)$ means there is $h \in H_W$ with $\langle e, Tg\rangle_E = \langle h, g\rangle_{H_W}$ for every $g \in D(T)$. Substituting $g = Uf$ gives the representing vector $U^{-1}h$ for T_Q^*, and conversely every $g \in D(T)$ is Uf for some $f \in D(T_Q)$. This proves the domain equality; density makes the representing vectors unique and proves the value formula. The codomain E is unchanged in this input transport, which is why its adjoint domain is literally the same subset of E.

As a human graph-norm reformulation, if $\mathcal{C}$ is a core for a closed T, then $U^{-1}\mathcal{C}$ is a core for T_Q: graph approximation of Uf pulls back to graph approximation of f, since both the input and output errors have exactly the same norm. This defines a transported closed calculus once an original operator is supplied. It does not identify T_Q with the sticky physical generator or a physical-time derivative. Nor does an isometry of complete path spaces identify their time-indexed filtrations, transition semigroups, or a prescribed SDE driver.

Source correspondence: `StickyWienerPathRealization`, `StickyBrownianRecovery`, `AmbientNullVersions`.
`StickyWienerPathEquivalence`, `StickyWienerL2Equivalence`.
`AdjointUnitaryTransport.mem_pullback_adjoint_domain`, `pullback_adjoint_eq`.

15 Two-time operators and the exact composition residual

Fix κ and the actual starting law of the constructed path. Let μ_t be the law of X_t^κ, let $H_t = L^2(\mu_t;\mathbb{R})$, and let $H = L^2(\mathbb{P};\mathbb{R})$ on the underlying source probability space. Pullback is an isometry

$$J_t : H_t \longrightarrow H, \qquad J_t f = f(X_t^\kappa).$$

Its range is exactly $L^2(\sigma(X_t^\kappa))$: one inclusion is immediate, and Doob–Dynkin applied to an almost-everywhere measurable representative proves the converse. The factor belongs to H_t by the pushforward norm identity. Thus the orthogonal projection $Q_t = J_t J_t^*$ is conditional expectation onto the *current state*, rather than the full observed past.

`BrownianStickyL2Transitions.embedding_range`.

`BrownianStickyL2Transitions.current_projection_condExp`.

Define $P_{s,t} = J_s^* J_t : H_t \to H_s$. For $f \in H_t$ and $g \in H_s$, its defining pairing is

$$\langle P_{s,t} f, g\rangle_{H_s} = \mathbb{E}[f(X_t^\kappa) g(X_s^\kappa)].$$

Conditional Jensen or the orthogonal projection inequality proves $\|P_{s,t} f\|_{H_s} \le \|f\|_{H_t}$. Its pullback is $\mathbb{E}[f(X_t^\kappa) \mid X_s^\kappa]$. It preserves nonnegativity and constants, $P_{t,t} = I$, and $P_{s,t}^* = P_{t,s}$. The last identity is the adjoint relation between two different marginal spaces, not stationarity or reversibility in one invariant marginal. The actual observed martingale theorem and the tower property also give $P_{s,t}(y \mapsto y) = (x \mapsto x)$ for $s \le t$; the coordinates lie in these spaces by the established second moment bound.

`BrownianStickyL2MarkovProperties.transition_coordinate`.

Since the state space is standard Borel, the actual pair law has a regular conditional probability kernel $K_{s,t}(x, dy)$. Disintegration gives $\mu_s(dx)K_{s,t}(x, dy)$ equal to that pair law and $\mu_s K_{s,t} = \mu_t$. Moreover $P_{s,t} f(x) = \int f(y) K_{s,t}(x, dy)$ for μ_s-almost every x. This is legitimate for $f \in H_t$: it is in $L^1(\mu_t)$, and disintegration first proves row integrability almost everywhere and then the conditional identity. Values of a chosen kernel at μ_s-null states are not determined by the pair law.

`BrownianTimeChangedConditionalKernel.pairLaw_disintegration`.

`BrownianStickyL2Transitions.transition_kernel`.

For every $f \in H_u$ there is the exact identity

$$P_{s,u} f - P_{s,t} P_{t,u} f = J_s^* (I - Q_t) J_u f. \tag{50}$$

This follows simply by expanding $J_s^* J_u f - J_s^* J_t J_t^* J_u f$. Thus composition holds on a particular test precisely when the current-state residual $(I - Q_t)J_u f$ is orthogonal to the earlier current-state subspace $J_s H_s$. It need not vanish in all of H. The human bound

$$\|P_{s,u} f - P_{s,t} P_{t,u} f\|_{H_s} \le \|(I - Q_t) J_u f\|_H$$

follows from $\|J_s^*\| \le 1$.

`BrownianStickyL2Transitions.composition_defect`.

`BrownianStickyL2Transitions.composition_iff_residual_zero`.

A full-past Markov identity would make this residual orthogonal to the entire past at t, and hence to $J_s H_s$ for $s \le t$. No such identity has been assumed to obtain (50). The whole-path isometry in the previous section likewise supplies no nesting of the one-time subspaces.

A finite martingale diagnostic (human example)

Even a martingale coordinate and all the positive two-time contractions above do not alone prove composition. Here is an explicit example, unrelated to a claim about the actual sticky path. Let ε, η, ζ be independent symmetric signs and set

$$Y_1 = \varepsilon, \qquad Y_2 = \varepsilon + \eta, \qquad Y_3 = Y_2 + a(\varepsilon)\zeta, \qquad a(1) = 1, \quad a(-1) = 2.$$

These bounded random variables form a martingale in the successive source filtration, and hence in their observed filtration, by independence and the zero mean of the fresh signs. Their conditional operators, defined exactly by $J_s^* J_t$, are positive contractions, preserve constants and the martingale coordinate, and have reverse adjoints. Yet composition fails. Take $f(y) = y^2$, an L^2 test for the finite marginal at time three. Conditional on $Y_1 = 1$, Y_2 is equally likely 0 or 2 and the final added variance is one. Hence

$$(P_{1,3} f)(1) = 3.$$

Conditional on $Y_2 = 2$, $\varepsilon = 1$ and the final square mean is 5. Conditional on $Y_2 = 0$, the two values of ε are equally likely, so the final square mean is $(1 + 4)/2 = 5/2$. Therefore

$$(P_{1,2} P_{2,3} f)(1) = \tfrac{1}{2}(5 + 5/2) = 15/4 \neq 3.$$

The defect is $-3/4$ at the positive-mass state 1. This elementary human counterexample is not a new named Lean declaration and is not a failure theorem for sticky Brownian motion. It identifies the logical role of the remaining Markov obligation: current-state conditional expectations do not form a nested tower merely because all coordinate martingale tests pass.

16 What is established and what remains separate

Clock recovery uses the observed prefix, not a terminal coordinate. It does not construct a strong solution with a prescribed physical-time Brownian driver. Bass's sticky SDE [2] distinguishes weak existence and uniqueness from strong existence; the operational representation (2) does not identify that driver filtration.

Section 13 reconstructs the densely defined closable continuous-function generator relation, its maximum-principle bound, and the explicit flat-test graph gap. Its range condition and closed-semigroup identification are not supplied by those results. The boundary martingales in Section 12 do not supply that missing range condition. No such missing step is assumed in this paper. In particular, positive two-time conditional operators alone do not establish Chapman–Kolmogorov consistency or an all-start transition family. No stochastic flow, generator-domain classification or uniqueness of the sticky martingale problem is asserted by Theorem 1.1.

The initial pair law gives an elementary version warning. Since $X_0^\kappa = 0$ almost surely, its marginal is $\mu_0 = \delta_0$ and the law of (X_0^κ, X_0^κ) is $\delta_{(0,0)}$. Consider the Borel probability kernels

$$K_1(x, \cdot) = \delta_x, \qquad K_2(x, \cdot) = \delta_0.$$

They agree μ_0-almost everywhere and both disintegrate that same pair law: for Borel A, E, either iterated measure gives $\mathbf{1}_A(0)\mathbf{1}_E(0)$. Nevertheless $K_1(1, \{0\}) = 0$ and $K_2(1, \{0\}) = 1$. Measurability follows because $K_1(x, E) = \mathbf{1}_E(x)$ and $K_2(x, E) = \mathbf{1}_E(0)$. Thus this actual initial pair law determines its conditional kernel only on states seen by μ_0. This example is not a pair of all-start transition semigroups: imposing $P_0(x, \cdot) = \delta_x$ for every initial state already excludes K_2 away from zero. It proves no nonuniqueness for a sticky martingale problem.

17 Formal provenance and reproducibility

All mathematical sources used here are frozen at

`bab688bb6140cd6316c80680095a914ea0d346db`.

The theorem map records exact declarations, complete headers, source hashes and baseline links for the mass, clock, martingale and pair constructions. The original full-build and transitive-axiom receipt and its referenced evidence hashes were checked. No mathematical source was changed and no competing research build was run.

The printed proofs reorganize verified arguments rather than adding formal declarations. Their distinctions between fixed-time L^2 classes, common continuous paths, raw-past versions and augmented actual coordinates are retained. The boundary martingale source is in the augmented physical past; its observed-past version here is an explicitly proved human filtration-shrinking deduction. The Hermite observed-past theorem is itself source-mapped. The pair proof supplies the full finite-increment argument beyond Gaussian marginals. The cited primary works provide classical antecedents; no new sticky process, local-time mechanism or priority is claimed.

References

[1] M. Ayala, G. Carinci and F. Redig, *Condensation of SIP particles and sticky Brownian motion*, Journal of Statistical Physics **183** (2021), article 40. doi:10.1007/s10955-021-02775-5.

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[3] C. Howitt and J. Warren, *Dynamics for the Brownian web and the erosion flow*, primary preprint (2007). arXiv:math/0702542.